

DEVELOPMENT OF THE PERIODIC TABLE

KEY VOCABULARY

Newlands: arranged elements in 'octaves' by atomic weight.

Mendeleev: left gaps and predicted undiscovered elements.

Atomic weight: the early ordering property (now atomic number).

Isotopes: explain why ordering by mass sometimes failed.

COMMON MISCONCEPTIONS

~~— Mendeleev had all the elements.~~

✓ He left gaps for elements not yet discovered.

~~— The table was always ordered by atomic number.~~

✓ Early tables used atomic weight.

~~— Newlands' table was fully accepted.~~

✓ It broke down and was criticised at the time.

MENDELEEV'S BREAKTHROUGH

Ordered elements by atomic weight (1869).

Left GAPS for undiscovered elements.

Swapped some pairs so properties matched the group.

Predicted properties of missing elements — later confirmed.

Ordering by atomic number (not weight) fixed the anomalies.

1 EARLY ORDERING

State the property early scientists used to **order** the elements. [1 mark]

2 MENDELEEV'S GAPS

Explain why Mendeleev left gaps in his table. [2 marks]

3 PREDICTIONS

Explain how the discovery of new elements gave Mendeleev's table credibility. [2 marks]

4 SWAPPED PAIRS

Some elements were placed out of strict mass order. **Explain** why. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Mendeleev ordered by atomic ____ but left ____ for missing elements.

gaps

weight

number

properties

6 THE MODERN TABLE

State what the modern periodic table is ordered by. [1 mark]